Aaradhya Dev Tamrakar

AI Safety & Interpretability · Formal Methods · Applied Machine Learning

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Final-year Electronics & Information Engineering undergraduate specializing in empirical AI safety, mechanistic interpretability, and formal verification of agentic systems. Fusemachines AI Fellow co-developing intersectional demographic fairness audit frameworks with statistical confidence intervals (AIF360, Fairlearn). Experienced in Z3 SMT invariant verification, Bayesian inference (PyMC/ArviZ), and quantized edge model evaluation.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems · Engineering Mathematics · Probability and Statistics · Object-Oriented Programming (C++) · Computer Networks · Microprocessors

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Engineered a five-stage agentic Text-to-SQL pipeline (Planner → Generator → Validator → Executor → Summarizer), achieving **100% execution success and answer accuracy** on a 50-question project benchmark.
- Developed customer churn and lifetime value (CLV) prediction pipelines (**5-fold CV ROC-AUC 0.841**) utilizing **SMOTE**, cost-sensitive threshold tuning, and **SHAP** explainability.
- Conducted Bayesian estimation and Gaussian-process regression with **PyMC/ArviZ**, and performed unsupervised segmentation (**K-Means**, **DBSCAN**) on **~500K retail transaction records**.

NSSR DataCamp Fellow (Cohort 2) — Nepalese Society of Student Researchers | April 2026 – Present

- Selected in competitive cohort for sponsored DataCamp access; developed six applied GenAI solutions automating data cleaning, schema normalization, and null filtering.

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026 – Present

PyTorch · Python · AIF360 · Fairlearn · FairFace · SHAP · Statistical Testing

- Co-developed an automated fairness-auditing framework with dual **AIF360** and **Fairlearn** backends computing **Disparate Impact Ratio (DIR)**, **Demographic Parity Difference (DPD)**, **Equal Opportunity Difference (EOD)**, and **Equalized Odds**.
- Audited classification performance across **126 granular race × age × gender groups** on the **FairFace benchmark (97,698 images)**, discovering significant disparity drop-offs across intersectional distribution shifts.
- Implemented rigorous statistical inference: **BCa bootstrap confidence intervals (B ≥ 1,000)**, **χ^2 and Fisher's exact tests**, **n ≥ 30 sample-size safeguards**, and **Shapley feature attribution (SHAP)** for flagged discrepancies.
- Architected clean CLI, HTML/PDF reporting pipelines, and modular test harnesses with **78/78 passing repository tests**; co-authored with Tisha Manandhar.

SPARK — Safety-Critical Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TFLite Micro · PyTorch · Python · BLE · FastAPI · SHAP

- Trained 1D-CNN on the **SisFall benchmark (38 subjects, 38,420 windows)**; achieved **0.9185 AUC-ROC (95% CI: [0.9065, 0.9305])** and **87.81% sensitivity** under subject-stratified 5-fold cross-validation with zero data leakage.

Headless Invariant Assurance Engine (INV-BMK-001) | 2026

Python · Z3 · SMT-LIB2 · State Machines · Pytest · Formal Methods

- Developed a formal verification benchmark encoding reactive state-machine contracts into **SMT-LIB2/Z3**, inductively proving candidate safety invariants and generating executable counterexamples.
- Evaluated **12 safety properties across 4 state machines**, discovering and sandbox-replaying **4/4 planted violations (100% recall, 0.0% false-discovery rate)** with **10.844 ms mean solver latency**.

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Direct branch strategy, executive committee operations, technical workshops, and peer learning initiatives for engineering students.
- Previously managed operational documentation and workshop logistics across department-level technical events as Vice Secretary.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL (PostgreSQL, SQLite), Kotlin, C# (.NET), VHDL, PowerShell, Bash, JavaScript / TypeScript

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, PyMC, ArviZ, SHAP, Fairlearn, AIF360, Hugging Face Transformers, OpenCV, MediaPipe

Formal Methods & Verification: Z3 SMT Solver, SMT-LIB2, State-Machine Contract Verification, Inductive Invariant Proofs, Counterexample Generation & Sandbox Replay

Engineering Practice & Tools: Git, GitHub, Linux (Ubuntu), Jupyter, Google Colab, LaTeX / Overleaf, WordCraft, Pytest, CMake / CTest, Automated CI/CD

CERTIFICATIONS, WORKSHOPS & ACTIVITIES

IEEE WIE Nepal LaTeX Workshop (May 2026) • IEEE SPAx (Student Professional Awareness Venture) Organizing Committee (2025 – 2026) • IEEEXtreme 19.0 Programming Competition (2025) • PCB Design & Fabrication Workshop (2025) • AWS Cloud Computing Workshop (2024) • Prompt Engineering Fundamentals (2024) • Microsoft Learn Student Ambassador (2024 – 2025)

ACADEMIC & PROFESSIONAL REFERENCES

Er. Dipen Manandhar — Faculty Member & Project Supervisor, Department of Electronics and Computer Engineering

Kathmandu Engineering College (IOE, Tribhuvan University). Context: Supervisor for undergraduate final-year SPARK major capstone project (2025–2026).

Shreejan Kisee — Teaching Assistant & Capstone Supervisor

Fusemachines AI Fellowship. Context: Supervised BiasAperture demographic fairness auditing research capstone and applied ML coursework (2026).